AKTIVITAS MANDIRI - LARAVEL

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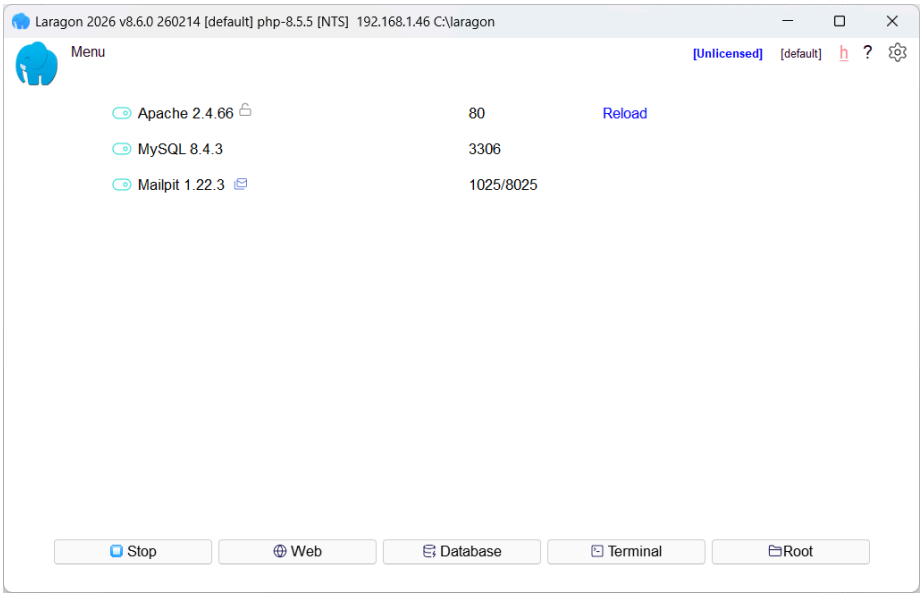
AKTIVITAS MANDIRI 1 - Instalasi dan Setup Proyek Laravel

1. Instal XAMPP terbaru, pastikan PHP 8.2+ tersedia (cek: php --version di terminal)



Disini saya menggunakan laragon untuk opsi alternatif XAMPP dan versi php adalah sebagai berikut:

```
PS C:\laragon\www> php --version
PHP 8.5.5 (cli) (built: Apr  7 2026 19:24:35) (NTS Visual C++ 2022 x64)
Copyright (c) The PHP Group
Built by The PHP Group
Zend Engine v4.5.5, Copyright (c) Zend Technologies
    with Zend OPcache v8.5.5, Copyright (c), by Zend Technologies
```

2. Instal Composer dari getcomposer.org, verifikasi dengan: composer --version
   Di komputer saya, sudah menginstall composer. Jadi ini adalah versi dari composernya:

```
PS C:\laragon\www> composer --version
Composer version 2.9.7 2026-04-14 13:31:52
PHP version 8.5.5 (C:\laragon\bin\php\php-8.5.5-nts-Win32-vs17-x64\php.exe)
Run the "diagnose" command to get more detailed diagnostics output.
```

3. Buat proyek Laravel baru: composer create-project laravel/laravel siak-app
   Karena saya sudah menginstall laravel installer di global composer, jadi saya hanya perlu menjalankan *laravel new pitstop* (pitstop adalah nama web saya).

```
PS C:\laragon\www> laravel new pitstop

Creating a "laravel/laravel" project at "./pitstop"
Installing laravel/laravel (v13.3.0)
  - Installing laravel/laravel (v13.3.0): Extracting archive
Created project in C:\laragon\www\pitstop
Loading composer repositories with package information
Updating dependencies
Lock file operations: 106 installs, 0 updates, 0 removals
> @php -r "file_exists('.env') || copy('.env.example', '.env');"

  INFO  Application key set successfully.


 Which database will your application use? [SQLite]:
  [sqlite ] SQLite
  [mysql  ] MySQL
  [mariadb] MariaDB
  [pgsql  ] PostgreSQL (Missing PDO extension)
  [sqlsrv ] SQL Server (Missing PDO extension)
 > mysql
mysql


 Default database updated. Would you like to run the default database migrations? (yes/no) [yes]:
 >


  INFO  Nothing to migrate.

 Would you like to run npm install --ignore-scripts and npm run build? (yes/no) [yes]:
 >


added 63 packages, and audited 64 packages in 10s

16 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities

> build
> vite build

vite v8.0.10 building client environment for production...
✓ 3 modules transformed.
computing gzip size...
public/build/manifest.json              0.33 kB │ gzip: 0.16 kB
public/build/assets/app-D05hqmaw.css   37.95 kB │ gzip: 9.38 kB
public/build/assets/app-34mOoJaZ.js     0.00 kB │ gzip: 0.02 kB

✓ built in 443ms

 Application ready in [pitstop]. You can start your local development using:

 → cd pitstop
 → composer run dev

 New to Laravel? Check out our documentation. Build something amazing!
```

4. Buat database baru siak_db di phpMyAdmin
   Membuat database dengan nama “pitstop” di phpMyAdmin

5. Konfigurasi file .env: isi DB_DATABASE, DB_USERNAME, DB_PASSWORD sesuai lokal

```
pitstop > .env
23  DB_CONNECTION=mysql
24  DB_HOST=127.0.0.1
25  DB_PORT=3306
26  DB_DATABASE=pitstop
27  DB_USERNAME=root
28  DB_PASSWORD=
29
```

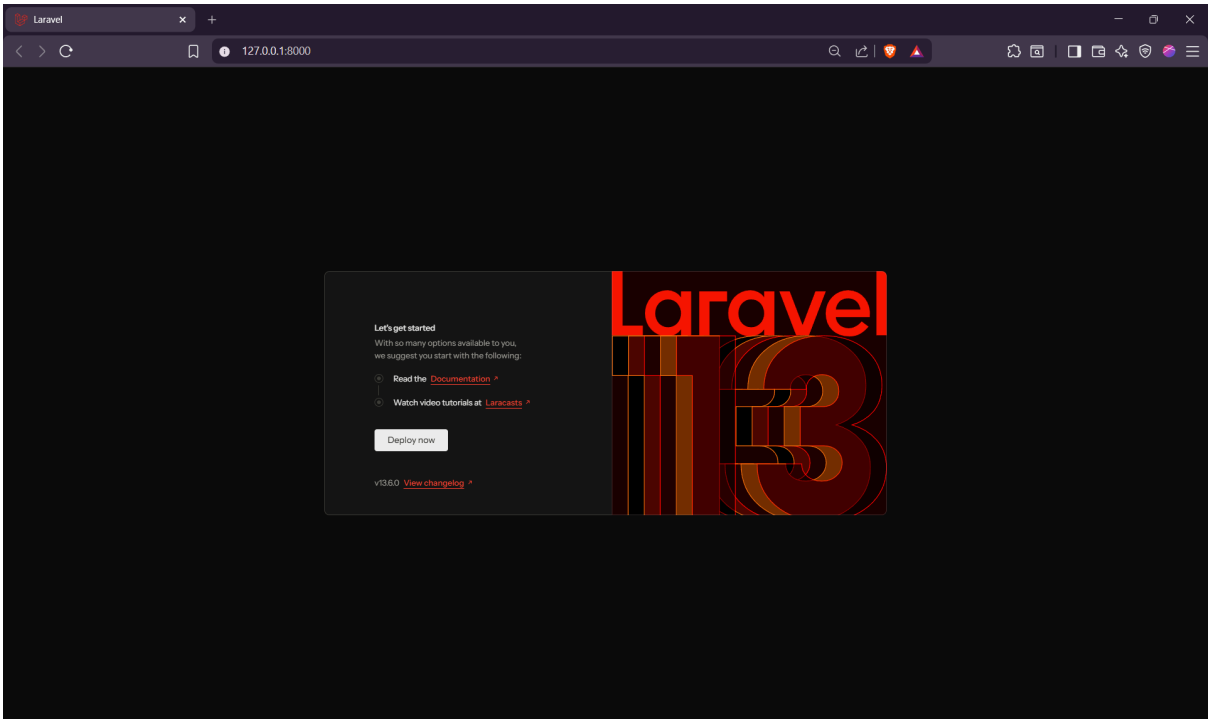
6. Jalankan: php artisan key:generate

```
PS C:\laragon\www\242410101063-PWEB> php artisan key:generate

  INFO  Application key set successfully.
```

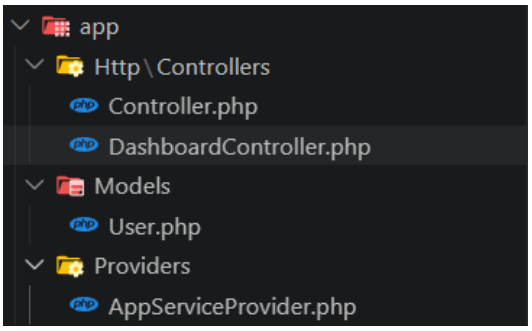
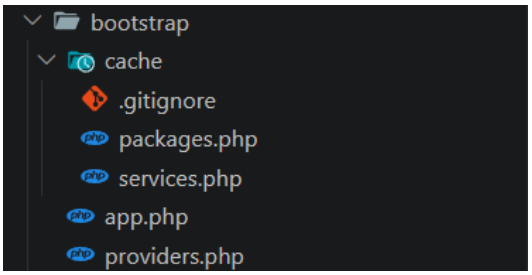
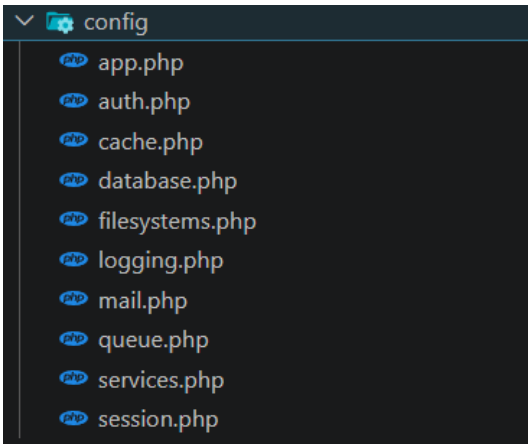
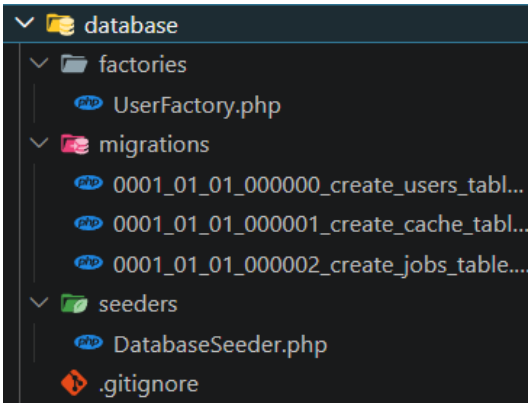
7. Jalankan server: php artisan serve - Lalu buka http://127.0.0.1:8000 di browser

```
PS C:\laragon\www\242410101063-PWEB> php artisan serve

  INFO  Server running on [http://127.0.0.1:8000].

  Press Ctrl+C to stop the server
```

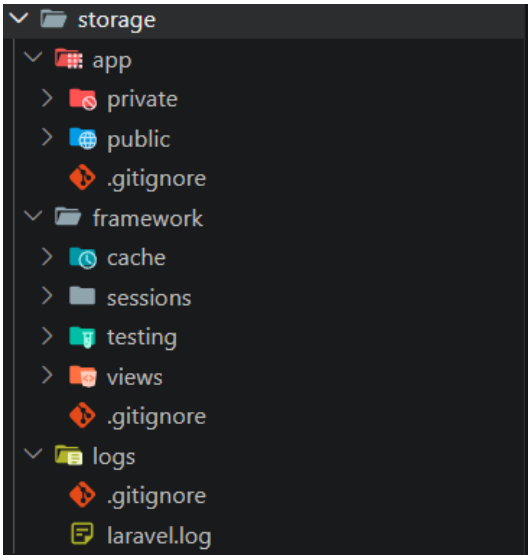
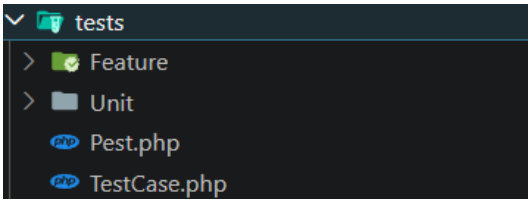
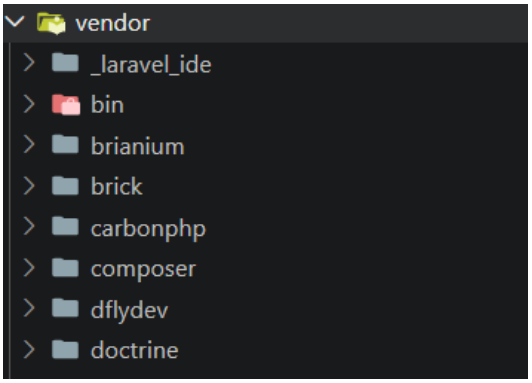
Lalu tampilan web awalnya:



8. Eksplorasi struktur folder, buka setiap folder dan pahami fungsinya

Folder	Fungsi
	Folder utama yang berisi logika inti aplikasi, seperti controller (mengatur alur request), model (interaksi database), dan service provider (konfigurasi global).
	Digunakan untuk proses inisialisasi Laravel dan menyimpan cache konfigurasi agar aplikasi berjalan lebih cepat.
	Berisi konfigurasi aplikasi seperti database, autentikasi, cache, mail, session, dan lainnya.
	Digunakan untuk mengatur database: struktur tabel (migration), data awal (seeder), dan data dummy (factory).

node_modules .bin .vite-temp @emnapi @jridgewell @napi-rs @oxc-project @rolldown @tailwindcss @tybys ansi-regex ansi-styles chalk cliui color-convert color-name concurrently detect-libc emoji-regex	Berisi dependency frontend dari npm (JavaScript, CSS, dll). Tidak perlu diubah manual.
public build assets app-D0Sx9e-A.css app-izHpl9rH.js manifest.json img favicon-dark.ico favicon-light.ico pitstop-logo.png .htaccess favicon.ico index.php	Folder yang bisa diakses langsung oleh browser, berisi file publik seperti CSS, JS, gambar, dan entry point aplikasi.
resources css app.css 3 js app.js views dashboard.blade.php 4	Berisi file mentah seperti Blade (tampilan), CSS, dan JavaScript sebelum diproses (build).
routes console.php web.php	Tempat mendefinisikan route aplikasi, baik untuk web maupun console.

	Digunakan untuk menyimpan file hasil proses aplikasi seperti log, cache, session, dan file upload.
	Digunakan untuk testing aplikasi (unit test dan feature test).
	Berisi library/dependency dari Composer (backend). Dan ini tidak boleh diubah manual.

.editorconfig .env .env.example .gitattributes .gitignore .npmrc artisan composer.json composer.lock package-lock.json package.json phpunit.xml README.md vite.config.js	File konfigurasi utama proyek seperti environment, dependency, dan command Laravel (artisan).

9. Jalankan: php artisan route:list

```
PS C:\laragon\www\242410101063-PWEB> php artisan route:list

  GET|HEAD  / ..................................................................... routes/web.php:5
  GET|HEAD  storage/{path} storage.local › vendor/laravel/framework/src/Illuminate/Filesystem/Filesy…
  PUT       storage/{path} storage.local.upload › vendor/laravel/framework/src/Illuminate/Filesystem…
  GET|HEAD  up vendor/laravel/framework/src/Illuminate/Foundation/Configuration/ApplicationBuilder.p…

                                                                                  Showing [4] routes
```

Ini adalah router awal dari laravel.

Aktivitas Mandiri 2 - Membuat Routing & Controller Dashboard

1. Buat Controller DashboardController:
   php artisan make:controller DashboardController

```
PS C:\laragon\www\242410101063-PWEB\pitstop> php artisan make:controller DashboardController

  INFO  Controller [C:\laragon\www\242410101063-PWEB\pitstop\app\Http\Controllers\DashboardController.php]
 created successfully.
```

2. Tambahkan method index() yang mengembalikan view('dashboard') di DashboardController

```
2 references | 0 implementations
class DashboardController extends Controller
{
    1 reference | 0 overrides
    public function index()
    {
        return view('dashboard');
    }
}
```

3. Daftarkan route di routes/web.php:
   Route::get('/dashboard', [DashboardController::class, 'index'])->name('dashboard');

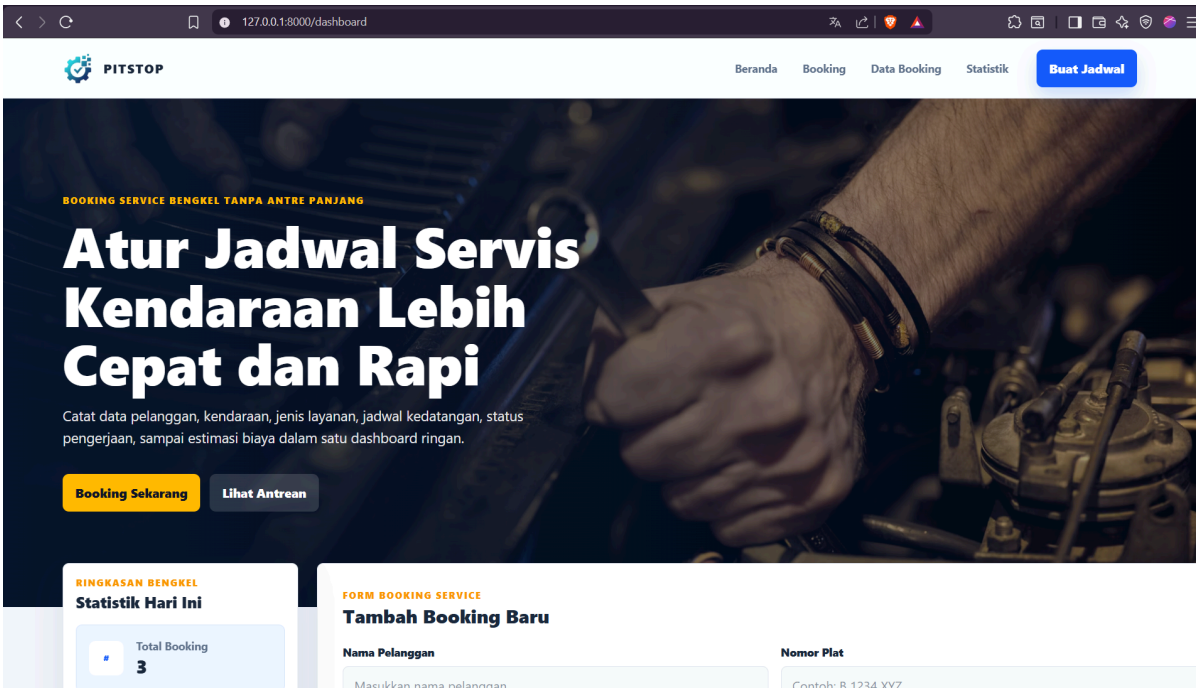
```php
<?php

use App\Http\Controllers\DashboardController;
use Illuminate\Support\Facades\Route;

Route::get('/', function () {
    return view('index');
});

Route::get('/dashboard', [DashboardController::class, 'index'])->name('dashboard');
```

4. Buat file view sementara resources/views/dashboard.blade.php berisi: <h1>Dashboard SIAK</h1>

```
resources > views > dashboard.blade.php > ...
<!doctype html>
<html lang="id">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>PitStop - Booking Service Bengkel</title>
    <link rel="icon" href="{{ asset('img/favicon-dark.ico') }}" media="(prefers-color-scheme: dark)">
    <link rel="icon" href="{{ asset('img/favicon-light.ico') }}" media="(prefers-color-scheme: light)">
    @vite(['resources/css/app.css', 'resources/js/app.js'])
</head>

<body class="min-h-screen overflow-x-hidden bg-[#f3f6fb] text-slate-800 antialiased">
    <header
        class="sticky top-0 z-20 flex flex-wrap items-center justify-between gap-4 border-b border-slate-200/90 bg-white/95 px-
        <a class="inline-flex items-center gap-3 font-black tracking-[0.08em]" href="#">
            <img src="{{ asset('img/pitstop-logo.png') }}" alt="Logo PitStop" class="h-10 w-10">
            <span>PITSTOP</span>
        </a>
```

Disini saya membuat dashboard yang sesuai dengan lanjutan projek kemarin.

5. Akses http://127.0.0.1:8000/dashboard di browser — pastikan tampil



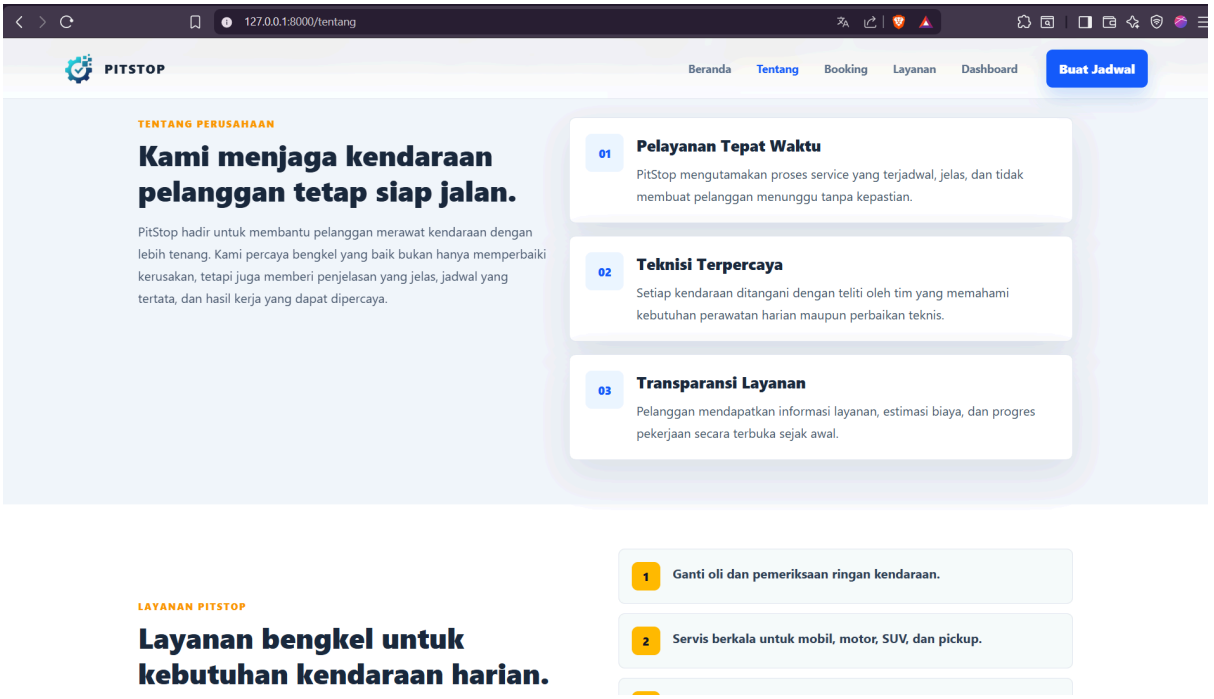
Di route ini sudah bisa tertampilkan.

6. Buat route /tentang dengan: Route::view('/tentang', 'tentang'); dan buat view-nya

```php
Route::view('/tentang', 'tentang');
```

```
resources > views > tentang.blade.php > ...
<!doctype html>
<html lang="id">

<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Tentang Kami - PitStop</title>
    <link rel="icon" href="{{ asset('img/favicon-dark.ico') }}" media="(prefers-color-scheme: dark)">
    <link rel="icon" href="{{ asset('img/favicon-light.ico') }}" media="(prefers-color-scheme: light)">
    @vite(['resources/css/app.css'])
</head>

<body class="min-h-screen overflow-x-hidden bg-[#f3f6fb] text-slate-800 antialiased">
    <header
        class="sticky top-0 z-20 flex flex-wrap items-center justify-between gap-4 border-b border-slate-200/90 bg-white/95 px-
        <a class="inline-flex items-center gap-3 font-black tracking-[0.08em]" href="{{ url('/') }}">
            <img src="{{ asset('img/pitstop-logo.png') }}" alt="Logo PitStop" class="h-10 w-10">
            <span>PITSTOP</span>
        </a>

        <input type="checkbox" id="menu-toggle" class="peer sr-only">
        <label for="menu-toggle"
            class="grid h-11 w-11 cursor-pointer place-items-center rounded-lg border border-slate-200 text-2xl leading-none te
            aria-label="Buka menu">
            &#9776;
        </label>

        <nav
            class="hidden w-full border-t border-slate-200 bg-white pt-4 shadow-[0_18px_45px_rgba(14,43,82,0.10)] peer-checked:
            <ul class="grid gap-4 text-sm font-bold text-slate-500 lg:flex lg:items-center lg:gap-8">
                <li><a class="transition hover:text-blue-600" href="{{ url('/') }}">Beranda</a></li>
                <li><a class="text-blue-600" href="{{ url('/tentang') }}">Tentang</a></li>
```

potongan kode view lanjutan dari projek kemarin.



7. Jalankan: php artisan route:list dan amati semua route yang terdaftar

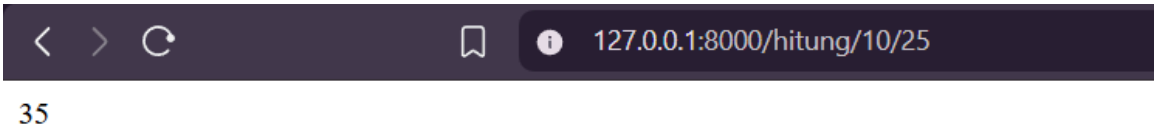
```
PS C:\laragon\www\242410101063-PWEB\pitstop> php artisan route:list

  GET|HEAD  / ..................................................... routes/web.php:6
  GET|HEAD  dashboard ............................... dashboard › DashboardController@index
  GET|HEAD  storage/{path} storage.local › vendor/laravel/framework/src/Illuminate/Filesys…
  PUT       storage/{path} storage.local.upload › vendor/laravel/framework/src/Illuminate/…
  GET|HEAD  tentang ...............................................................
  GET|HEAD  up vendor/laravel/framework/src/Illuminate/Foundation/Configuration/Applicatio…

                                                                  Showing [6] routes
```

Disini route-route tadi (dashboard, tentang) sudah terdaftar.

8. Coba buat route: Route::get('/hitung/{a}/{b}', fn($a, $b) => $a + $b); lalu akses /hitung/10/25
   Pada proyek saya sebenarnya tidak ada route yang seperti ini, jadi route ini hanya saya buat untuk percobaan dan tidak berhubungan dengan aplikasi utama.
   Ini routenya:

```
Route::get('/hitung/{a}/{b}', fn ($a, $b) => $a + $b);
```

   Lalu, ini jika kita akses route ini:



   Jadi di halaman web, ini akan menjalankan fungsi yang terdaftar di route. Tujuan dari aktivitas ini kemungkinan untuk memahami bagaimana cara menangani (handle) request GET dari URL yang telah terdaftar di dalam route.